HR Feedback Intelligence

Streamlit application for explainable HR feed back analysis and 9-box talent category prediction developed for the Explainability AI hackathon.

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The Team

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Project Objectives

Actionable Insights

Transform raw comments into readable trends for HR and managers.

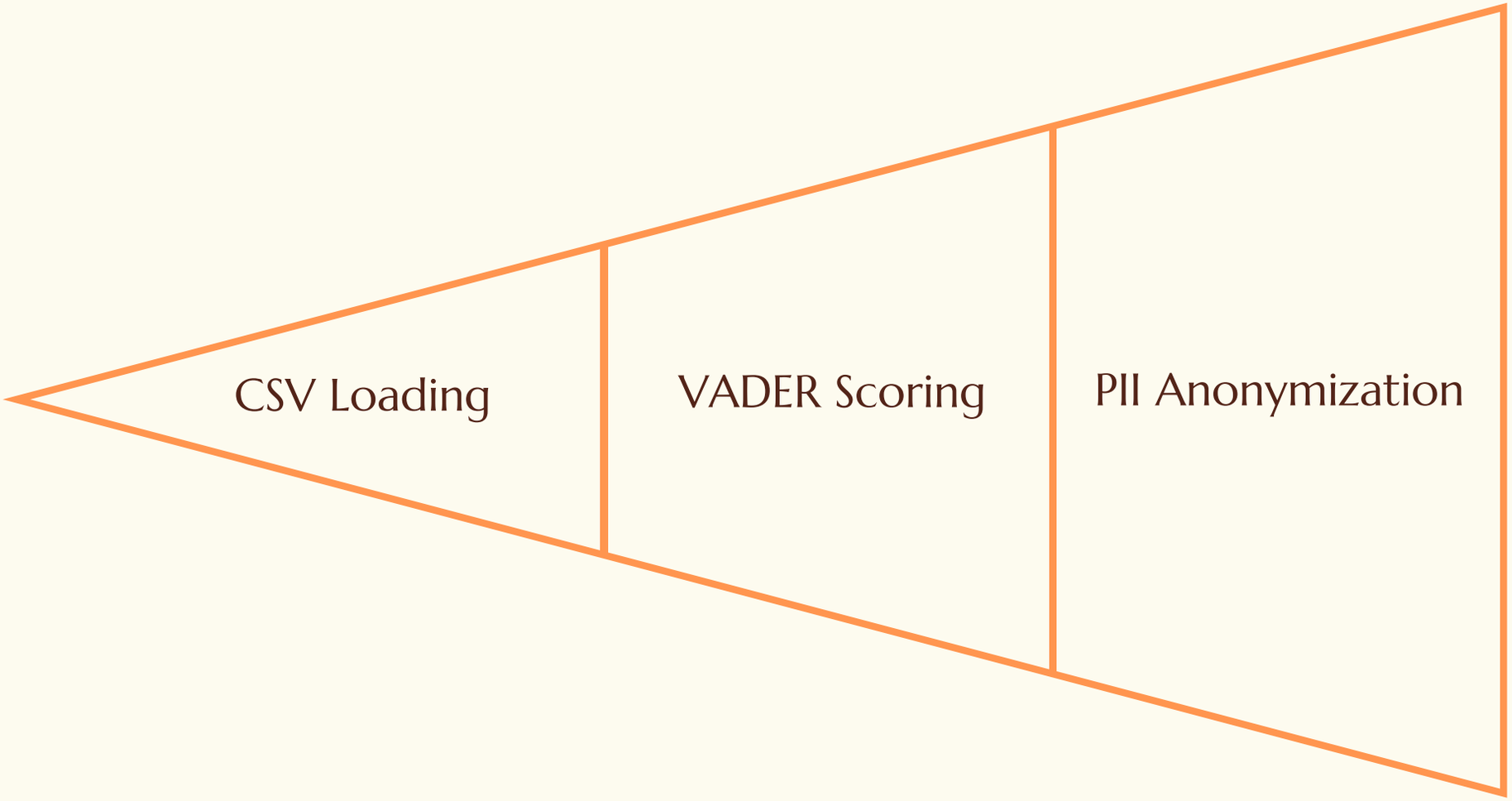
Transparency

Explainable outputs (charts, counts) rather than black-box decisions.

9-Box Prediction

Predict an employee's talent category based on feedback text.

Processing Pipeline



The business logic pipeline processes each feedback comment from end to end, from ingestion to enriched export, without manual intervention.

Interface Modules

Summary KPIs
Entries, average sentiment, counts

Sentiment Visuals
Pie chart + histogram

Explorer
Filtering and comment navigation

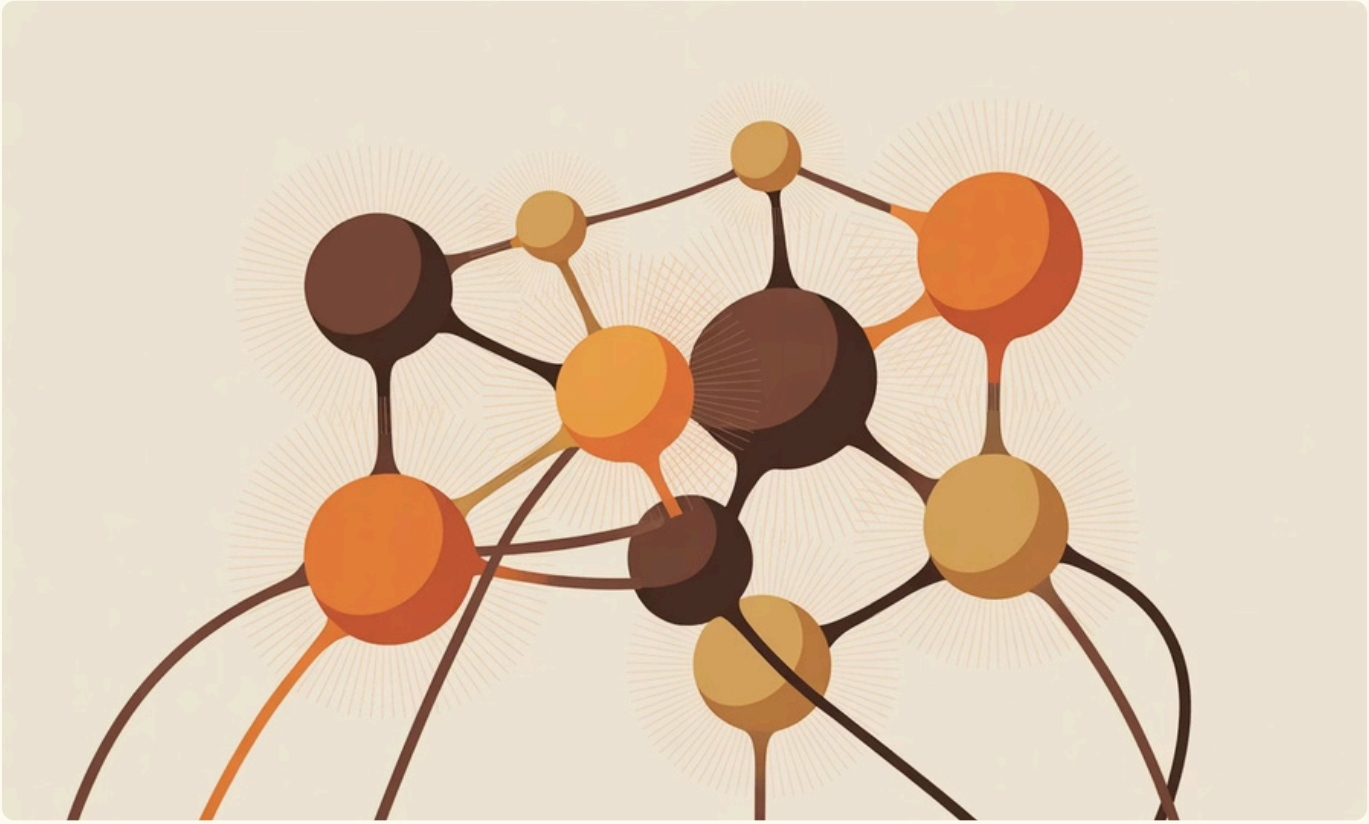
9-Box Heatmap
Sentiment by talent category

Diagnostics
Topics, bias, quality & coherence

BERT Section
Metrics, confusion matrix, sandbox

CSV Export
Download enriched analysis

DistilBERT Model — Configuration



Key Parameters

Base Model

distilber t-base -uncased

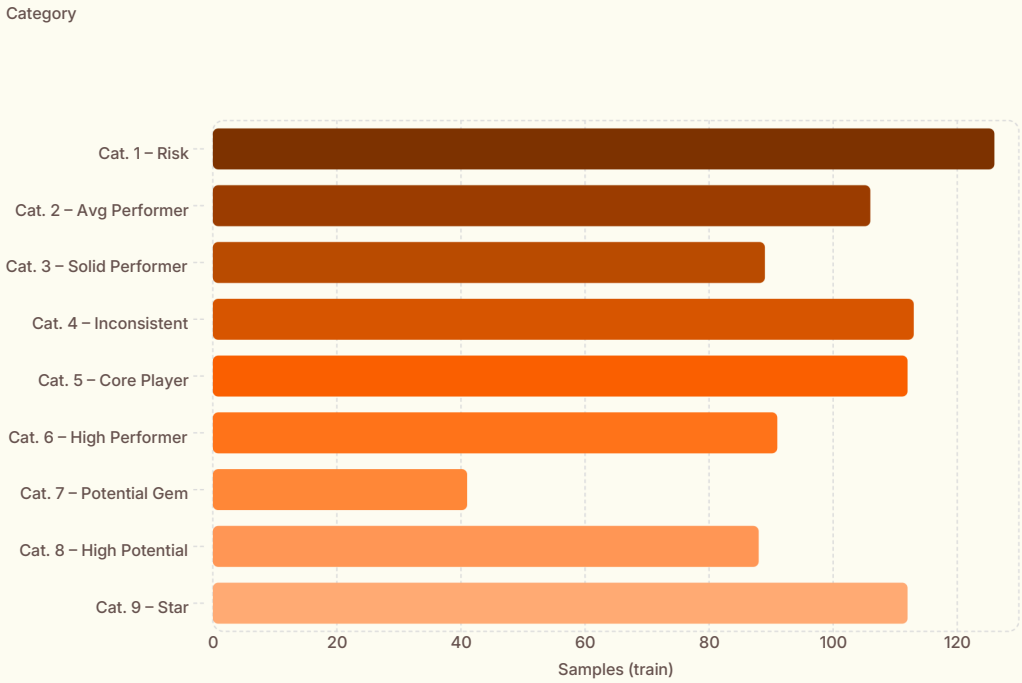
Task

9-class classification
(nine_box_category)

Hyperparameters

max_len=128 · batch=16 ·
epochs=3 · lr=2e-5

Training Data



Dataset

1,103 samples in total — 878 training, 225 test (25 per class, balanced).

Source: Kaggle (local copy). Varied HR comments on performance and potential.

⚠ Category 7 (Potential Gem) is underrepresented with only 41 examples.

Model Performance

37.78%	0.31	256 MB	11.57 ms
Accuracy	Macro F1	Model Size	CPU Inference
On the test set	Macro-averaged F1 score	Artifacts saved to disk	Average time per comment

Confusions mainly occur between neighboring classes in the 9-box matrix. Generalization depends on the quality and size of the dataset.

Risks & Limitations

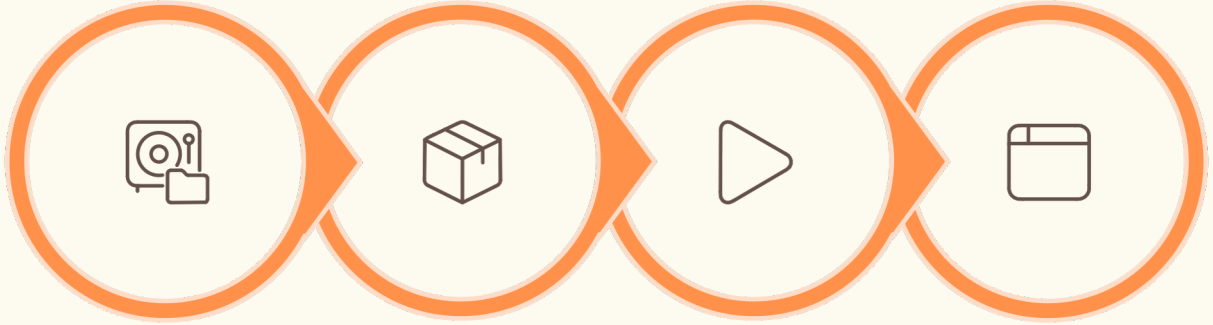
Known Limitations

- Business-rule signals are insensitive to context and sarcasm
- Proxy bias via pronouns: heuristic only
- Slow CPU training on low-power machines
- Out-of-distribution data (other languages, styles) not covered

Control Measures

- In-app warning message: group usage, not automated individual decisions
- Transparent display of metrics
- Mandatory human review workflow
- CSV export for audit
- Anonymization at display

Quick Start



Load your data (integrated dataset or personal CSV), explore the analyses, run DistilBERT, and export the enriched results.